A Deep Cross-Modal Fusion Network for Road Extraction With High-Resolution Imagery and LiDAR Data

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***Abstract*— Urban road extraction is important for the applications of urban planning and transportation. High-resolution image (HRI) has been one of the most popular data sources for extracting roads with high efficiency and low cost. However, roads in HRI are easily obscured by buildings, trees, and other landscapes, resulting in discontinuity of the extracted roads. While current road extraction techniques by multimodal data fusion have shown improved results compared to single-modal methods by incorporating additional information, most existing fusion methods fail to fully exploit the features from different modalities and consider prior knowledge of roads. To address the above problems, a dual encoder-based cross-modal complementary fusion network (DECCFNet) is proposed in this article. The proposed network takes full advantage of the rich feature information contained in HRI and the immunity of LiDAR data to the influence of shadows. By effectively fusing the complementary information from HRI and LiDAR data, DECCFNet, respectively, achieved an improvement by at least 2.94% and 2.8% in IOU compared to those only using a single data modality on the two datasets. The proposed DECCFNet mainly contains two modules: 1) cross-modal feature fusion (CMFF) module: in the dual encoder part, CMFF is employed to fuse the deep features of different modalities from the channel and spatial dimension, while a multiscale fusion strategy is utilized to extract the contextual information; 2) multi-direction strip convolution (MDSC) module: since roads have the characteristics of narrowness and continuity, adopting classical convolution kernels directly on road features may introduce irrelevant pixels into the computation, blurring the extraction results. To mitigate this issue, MDSC is applied to strip the convolution of road features from multiple directions based on square convolution and make the network focus more on the specific road features. By comparing several deep-learning multimodal data fusion networks in the two road datasets, the proposed network exhibits the best road extraction results.**

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***Index Terms*— Convolutional neural network, cross-modal feature fusion (CMFF), high-resolution remote sensing image, LiDAR data, road extraction.**

I. Introduction

URBAN road network is one of the most important geographic information data which plays an irreplaceable role in various applications, such as automatic navigation, unmanned vehicles, urban planning and transportation, and geographic information updating [1], [2], [3], [4]. Thus, accurately and automatically obtaining roadway data is significant and worthwhile. With the continuous improvement of remote sensing technology, we can obtain a huge amount of different types of remote sensing data, and they have become the main data source of road extraction.

Currently, the commonly used remote sensing data for road extraction include high-resolution images (HRIs), airborne LiDAR point cloud data, SAR images, and hyperspectral data. Among them, high-resolution images show the advantages of being easily accessible and containing rich detailed information. Road extraction from high-resolution images using deep convolutional networks has become a popular research direction [5], [6], [7], [8], [9]. However, roads in high-resolution images may have visual discontinuities due to obscure trees and buildings, and road extraction is also negatively affected by the different scales of roads and similar features of other landscapes. Therefore, researchers have improved the road extraction results by expanding the receptive field, aggregating multiscale features, and introducing attention mechanisms to mine the contextual information as much as possible [10], [11], [12], [13], [14], [15], [16], [17], [18]. Zhou et al. [19] proposed a road extraction network based on LinkNet structure, which introduced a cascade expansion convolution module in the top layer of the encoder as receptive fields in the center part of this network. Wan et al. [20] used dense connectivity layers to stack multilevel features to reduce road information loss, while designing an attention module to enhance road feature representation by weighted fusion of low-level detail features with high-level semantic features. Another way to improve the performance of road extraction is to incorporate spatial topological characteristics of roads to improve road continuity [21], [22]. Tao et al. [23] proposed a spatial information inference structure and combined it with recurrent neural network (RNN) units to make the contextual information

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propagate from along the road, which effectively maintains the continuity of the extracted road. Wu et al. [24] considered the topology of points, edges, and directions of roads and proposed a method for predicting bi-directionally connected roads based on the heat map of road nodes. Yuan and Xu [25] suggested the GapLoss attention loss function for road topology, which effectively extracted intersection roads and roads obscured by shadows. Although the above methods have tried their best to eliminate the influence of shadows and other similar features on road extraction in high-resolution images, these methods are unable to break through the data modality constraints relying only on a single data modality. Hence, multimodal data fusion road extraction techniques are gradually developed.

There are already a variety of road extraction techniques by the fusion of different modal remote sensing data, such as the fusion of SAR images [26], [27], [28] and hyperspectral images with other remote sensing data [29], [30], [31], [32]. Particularly, compared with high-resolution remote sensing images, LiDAR data are not affected by shadows, although the feature edge information is rougher. The processed LiDAR data can not only reflect the relative ground information of ground objects but also represent the material properties of different ground objects. Naturally, LiDAR data make up for the shadow effects and similar spectral features and confusion defects of high-resolution images, so the road extraction techniques based on the fusion of high-resolution images and LiDAR data develop more rapidly. Hu et al. [33] presented an urban road network extraction approach that combines LiDAR data and HRIs, and the above approach greatly improves the accuracy of road extraction compared to using only a single modality of HRI or LiDAR data. Zhang et al. [34] utilized LiDAR data and high-resolution images to extract road centerlines, and experiments conducted on multiple datasets proved that the effect of the road extraction method was still valid in complex scenarios. Parajuli et al. [35] formed a bimodal setup and fused RGB images and LiDAR depth data for extracting road networks through pixel-by-pixel segmentation, proposed multiple fusion strategies and demonstrated that data fusion was beneficial for improving road extraction accuracy. Subsequently Parajuli et al. [36] suggested a CNN network combination named QuadRoad, which used three modal remote sensing data, high-resolution aerial imagery, airborne LiDAR, and low-resolution multispectral imagery, to train multiple CNN models, and finally synthesized the prediction results of multiple CNN models to output the final road extraction results for improving the accuracy of binary road segmentation.

Meanwhile, semantic segmentation methods for remote sensing ground objects based on the fusion of high-resolution images and LiDAR data are also developing rapidly [37], [38], [39], [40], [41], [42]. Currently, the mainstream techniques utilize deep learning techniques to extract and fuse different modal deep features from multimodal data to enhance the semantic segmentation results [43]. Audebert et al. [44] investigated the effect of early and late fusion of high-resolution images and LiDAR data on semantic segmentation results based on deep convolutional networks. The experimental results showed that both fusion methods can effectively utilize the complementarity of different modalities to enhance the semantic segmentation results. Subsequently, Cao et al. [45] examined various fusion approaches based on DSM fusion branches to explore the best feature fusion method. However, most of their fusion strategies used simple summation or concatenation operations, ignoring the internal differences between modality features. Therefore, researchers have done many studies on the complementarity of multimodal feature differences. Sun et al. [46] employed attentional mechanisms and multimodal learning strategies to adaptively learn and integrate differential features across modalities. Zheng et al. [47] utilized cross-modal descriptors during the network to generate channel-guided weights for calibrating RGB features, in order to preserve target features while suppressing redundant and noisy information. Zhao et al. [48] exploited the collaborative fusion module and the enhanced aggregation module to address the intraclass differences and interclass similarities in features of different modal features. Ma et al. [49] took advantage of the transformer architecture and employed cross-attention mechanisms to merge dual-modality features across multiple scales, capturing long-distance dependencies in multiscale feature maps. Zhang et al. [50] used a hybrid fusion strategy to pay attention to individual modal and cross-modal features, in which the channel attention mechanism was utilized to adaptively reweight the different modal feature channels, and finally increased the building extraction accuracy. Hosseinpour et al. [51] proposed cross-modal and multilevel feature fusion methods to promote multimodal feature fusion and thus improve building extraction results.

The above methods demonstrate that the integration of high-resolution images and LiDAR data can enhance the accuracy of semantic segmentation. Nevertheless, existing road extraction techniques based on high-resolution images and LiDAR data overlook the complementary fusion of deep features between modality data when adopting deep learning techniques, and do not fully explore the deep semantic information of different modalities for roads. In addition, in remote-sensing semantic segmentation methods for the fusion of HRIs and LiDAR data, despite increased focus on differences and complementarities of multimodal deep features, the spatial and channel-wise information varies between features of different scales, and there is still a large research space for effectively integrating these complementary features. Furthermore, given the target of road extraction and the morphological characteristics of roads as being narrow and continuous, better feature extraction methods are needed to focus the network on road features, thereby further enhancing road extraction results.

To address the above challenges, this article proposes a dual encoder-based cross-modal data complementary fusion network (DECCFNet). DECCFNet is a U-shaped network with two encoders and one decoder. Dual encoders extract HRI and LiDAR modal features, respectively, and the cross-modal feature fusion (CMFF) module is proposed to rectify and integrate the two modal features extracted from each layer of the encoding block, then fused features are fed to the next encoding layer, while they are also sent to the corresponding decoding layer via skip connections to supplement the detail

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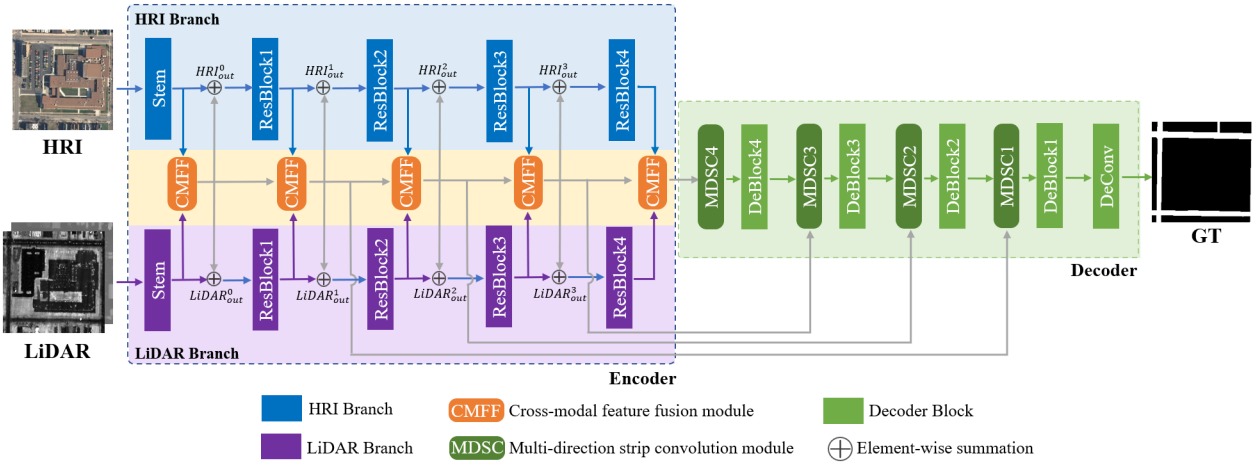

Fig. 1. Overall architecture of the DECCFNet.

information. To target the characteristics of roads that are narrow and not easily interrupted, a multi-direction strip convolution (MDSC) module is introduced in each decoding layer to convolve the features from multiple directions, which makes the network more conducive to focusing on the road features. In addition, a mixed loss function is implemented to train the network taking into account the highly imbalanced nature of the road dataset categories. The main contributions of this article are as follows.

1) A novel CMFF module is designed. Compared with the existing cross-modal fusion module for remote sensing ground object extraction, CMFF focuses on helping the network to effectively fuse multimodal deep features from channel and spatial dimensions and to ensure the advantageous features of each modality while learning complementary information from each other.

2) An MDSC module is introduced to convolve the features from multiple directions on the basis of square convolution for road characteristics, so as to enable the network more focused on road features and further improve road extraction accuracy.

3) A new DECCFNet for road extraction from HRIs and LiDAR data is proposed and extensively experimented on the Erie dataset and USGS dataset, and the experimental results prove the effectiveness and generalization of the proposed method.

II. Proposed Method

A. Network Structure

The proposed DECCFNet is a U-shaped framework with dual encoders and a decoder. DECCFNet takes ResNet-34 pretrained on ImageNet as the base encoder. The first 7×7 downsampling block of ResNet-34 is replaced by three 3×3 convolutional blocks to retain more detailed information. Therefore, the first layer of ResNet-34 pretraining weights is discarded, and the number of convolutional channels in the first layer is changed to the number of channels matching the training data. The Stem convolutional weights are initialized using kaiming. DECCFNet is shown in Fig. 1. At the encoder, the CMFF module rectifies and fuses the HRI features and LiDAR features extracted from each encoding block in the channel and spatial dimensions. The fused features are discriminatively input to the next encoding layer in the corresponding branches. Meanwhile, the fused features are fed into the next CMFF for multiscale fusion and added to the corresponding decoding layer for additional detailed information with skip connections. In the decoder, as roads are usually narrow and continuous, normal square convolution tends to introduce unnecessary information for computation, thus MDSC module is applied to convolve features from multiple directions to make the network focus on road features.

B. CMFF Module

Due to the fact that the information of different modalities is usually distinct and complementary, directly adding or stacking them will lead to information redundancy and bring in more noise. Therefore, we propose a new CMFF module, as shown in Fig. 2. At each feature extraction stage in the encoder, different modal features $F^i_{\text{HRI}}, F^i_{\text{LiDAR}} \in \mathbb{R}^{C_i \times H_i \times W_i}$ are rectified from the channel dimension and integrated from the spatial dimension for containing complementary information. The multiscale fusion strategy is utilized to obtain fusion features $F^i_{\text{fused}} \in R^{C_i \times H_i \times W_i}$. C_i, H_i, and W_i denote the number of channel, height, and width for the ith ($i = 0, 1, 2, 3, 4$) encoding layer feature. Inspired by SE block [52], the channel attention mechanism is adopted to enhance features. Specifically, we squeeze the input features F^i_{HRI} and F^i_{LiDAR} separately in the channel dimension to acquire the respective channel weights $C^i_{\text{HRI}}, C^i_{\text{LiDAR}} \in [0, 1]^{C_i \times 1 \times 1}$, and then multiply the weights with the original input features to obtain the enhanced features $\widetilde{F}^i_{\text{HRI}}$ and $\widetilde{F}^i_{\text{LiDAR}}$

$$C^i_{\text{HRI}} = \delta(\text{MLP}(\text{AvgPool}(F^i_{\text{HRI}})) + \text{MLP}(\text{MaxPool}(F^i_{\text{HRI}}))) \quad (1)$$

$$C^i_{\text{LiDAR}} = \delta(\text{MLP}(\text{AvgPool}(F^i_{\text{LiDAR}})) + \text{MLP}(\text{MaxPool}(F^i_{\text{LiDAR}}))) \quad (2)$$

$$\widetilde{F}^i_{\text{HRI}} = C^i_{\text{HRI}} \odot F^i_{\text{HRI}} \quad (3)$$

$$\widetilde{F}^i_{\text{LiDAR}} = C^i_{\text{LiDAR}} \odot F^i_{\text{LiDAR}} \quad (4)$$

where $\odot$ represents the channel-wise multiplication and δ denotes the Sigmoid activation function.

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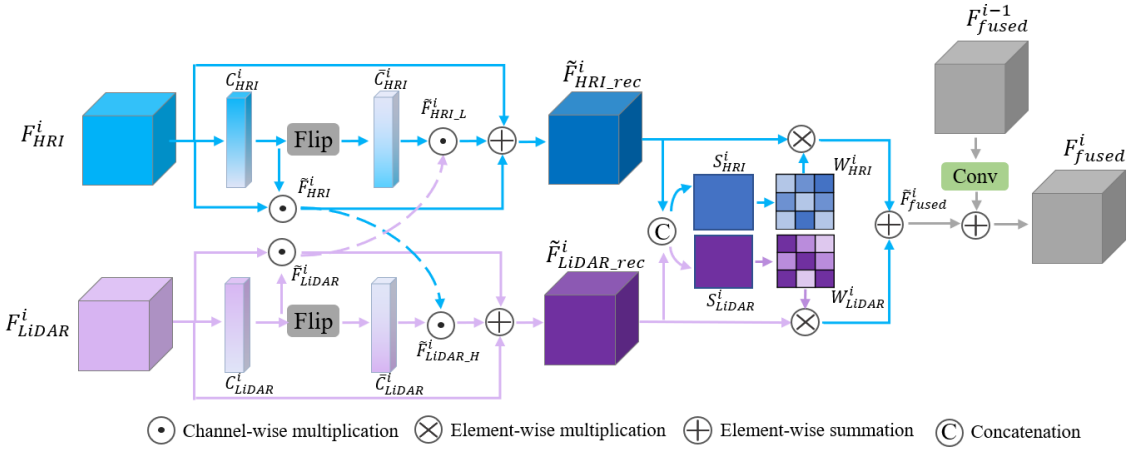

Fig. 2. Structure of CMFF module.

Taking HRI features as an instance, the channel weight $C^i_{\text{HRI}} \in [0,1]^{C_i\times 1\times 1}$ represents the significance of each channel in the features F^i_{HRI}. The higher weight means that the channel contains more valid information, which can be called the peak channel, while the opposite means that the channel contains less valid information, which can be named the low peak channel. The enhanced features of other modality feature $\widetilde{F}^i_{\text{LiDAR}}$ are multiplied with the weights of the low-peak channels $\overline{C}^i_{\text{HRI}}$ of HRI modality to gain the middle features. In this way, the valid information from other modalities is able to propagate into the present modality features to complement the information in the low peak channel, which achieves cross-modal feature complementarity, and the same is true for the LiDAR features

$$\widetilde{F}^i_{\text{HRI_}L} = \text{Conv}\left(\widetilde{F}^i_{\text{LiDAR}}\right) \odot \left(1 - C^i_{\text{HRI}}\right) \tag{5}$$

$$\widetilde{F}^i_{\text{LiDAR_}H} = \text{Conv}\left(\widetilde{F}^i_{\text{HRI}}\right) \odot \left(1 - C^i_{\text{LiDAR}}\right) \tag{6}$$

where Conv represents the 3×3 convolution. The final complementary rectified middle features of the two modalities $\widetilde{F}^i_{\text{HRI_rec}}$ and $\widetilde{F}^i_{\text{LiDAR_rec}}$ are as follows:

$$\widetilde{F}^i_{\text{HRI_rec}} = F^i_{\text{HRI}} + \widetilde{F}^i_{\text{HRI}} + \widetilde{F}^i_{\text{HRI_}L} \tag{7}$$

$$\widetilde{F}^i_{\text{LiDAR_rec}} = F^i_{\text{LiDAR}} + \widetilde{F}^i_{\text{LiDAR}} + \widetilde{F}^i_{\text{LiDAR_}H}. \tag{8}$$

In order to fully exploit the complementary properties of different modal features, we perform complementary integration of each modal feature from the spatial-wise dimension. We propose a spatial attention gate that allows two modal features $\widetilde{F}^i_{\text{HRI_rec}}$ and $\widetilde{F}^i_{\text{LiDAR_rec}} \in \mathbb{R}^{C_i\times H_i\times W_i}$ to exhibit complementary importance at the same location in space and reweights the spatial features of each modality, ultimately enabling spatial feature integration. Concretely, we stack two modal features $\widetilde{F}^i_{\text{HRI_rec}}, \widetilde{F}^i_{\text{LiDAR_rec}} \in \mathbb{R}^{C_i\times H_i\times W_i}$, map them to spatial attention gates $[S^i_{\text{HRI}}, S^i_{\text{LiDAR}}] \in \mathbb{R}^{1\times H_i\times W_i}$, and apply the Softmax activation function to obtain spatial position weights

$$\left(W^i_{\text{HRI}}, W^i_{\text{LiDAR}}\right) = \delta\left(\text{Conv}\left(\text{Cat}\left(\widetilde{F}^i_{\text{HRI_rec}}, \widetilde{F}^i_{\text{LiDAR_rec}}\right)\right)\right) \tag{9}$$

$$W^i_{\text{HRI}} + W^i_{\text{LiDAR}} = 1 \tag{10}$$

where $W^i_{\text{HRI}}, W^i_{\text{LiDAR}} \in \mathbb{R}^{1\times H_i\times W_i}$, and δ indicates the Softmax activation function, and Conv denotes 1×1 convolutional layer.

The input modal features $\widetilde{F}^i_{\text{HRI_rec}}$ and $\widetilde{F}^i_{\text{LiDAR_rec}}$ are multiplied with the corresponding spatial position weights to obtain the cross-modal complementary aggregated features $\widetilde{F}^i_{\text{fused}}$

$$\widetilde{F}^i_{\text{fused}} = \widetilde{F}^i_{\text{HRI_rec}} * W^i_{\text{HRI}} + \widetilde{F}^i_{\text{LiDAR_rec}} * W^i_{\text{LiDAR}}. \tag{11}$$

In order to better capture the contextual relationships of the cross-modal fusion features at different scales, we adopt a multiscale feature fusion strategy, in which the $i-1$th layer of fusion features F^{i-1}_{fused} are fused with $\widetilde{F}^i_{\text{fused}}$ in a multiscale fusion to obtain the final fusion features F^i_{fused}

$$F^i_{\text{fused}} = \begin{cases} \widetilde{F}^i_{\text{fused}}, & i = 0 \\ \widetilde{F}^i_{\text{fused}} + \text{Down}\left(\text{Conv}\left(F^{i-1}_{\text{fused}}\right)\right), & i = 1,2,3,4 \end{cases} \tag{12}$$

where Conv indicates the 1×1 convolutional layer used to change the number of channels and Down denotes the downsampling operation.

The fusion features F^i_{fused} are fed into HRI and LiDAR branches to complement the original ith layer features F^i_{HRI} and F^i_{LiDAR} with interactions between different modalities for inputs of next layer

$$\text{HRI}^i_{\text{out}} = \left(F^i_{\text{HRI}} + F^i_{\text{fused}}\right)/2 \tag{13}$$

$$\text{LiDAR}^i_{\text{out}} = \left(F^i_{\text{LiDAR}} + F^i_{\text{fused}}\right)/2. \tag{14}$$

C. MDSC Module

Currently, the majority of deep convolutional networks extract features through square convolutional kernels. On the contrary, roads are not like the common remote-sensing ground objects showing block shapes, such as buildings and trees, and they have the morphological characteristics of being thin and continuous without easy interruptions. Using traditional square convolution kernels probably introduces surrounding irrelevant pixels for computation, blurring the road features. Besides, roads have greater requirements for contextual information and long-distance dependencies. Motivated by Hou et al. [53] and Sun et al. [54], we propose an MDSC module based on road morphological features. As shown in Fig. 3, MDSC

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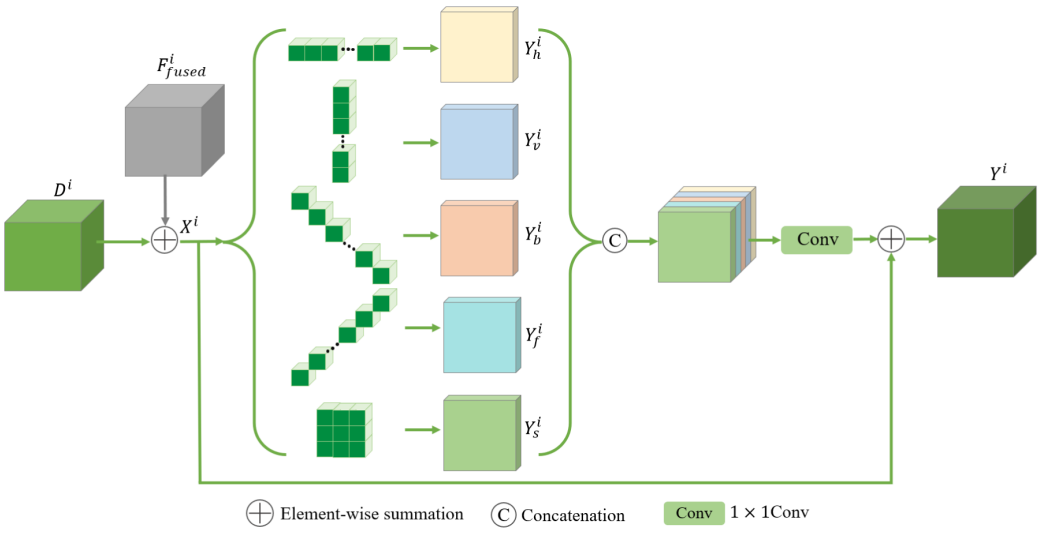

Fig. 3. Structure of MDSC module.

is based on square convolution and utilizes horizontal, vertical, front diagonal, and back diagonal strip convolution in multiple each direction for road feature extraction, which supplements the road detail information and focus the deep network more on the road features. Specifically, the input features $D^i \in \mathbb{R}^{C_i \times H_i \times W_i}$ are summed with the fusion features to get features $X^i \in \mathbb{R}^{C_i \times H_i \times W_i}$. Then, the features X^i are fed into five branches for convolution, each of which is a square convolution and a 1-D transposed convolution in the horizontal, vertical, front diagonal, and back diagonal directions, respectively. The intermediate features are obtained by stacking, then the final features $Y^i \in \mathbb{R}^{C_i \times H_i \times W_i}$ are acquired by summing the residual connections with the original input features which are finally input to the corresponding decoding layer, where C_i, H_i, and W_i denote the number of channel, height, and width for the ith ($i = 1, 2, 3, 4$) decoding layer feature

$$X^i = \begin{cases} D^i, & i = 4 \\ D^i + F^i_{\text{fused}}, & i = 1, 2, 3 \end{cases} \tag{15}$$

$$Y^i = X^i + \text{Conv}\left(\text{Cat}\left(Y^i_s, Y^i_h, Y^i_v, Y^i_f, Y^i_b\right)\right) \tag{16}$$

where Conv denotes the 1×1 convolution layer used to recover the number of channels, Y^i_s denotes the feature calculated by a 3×3 convolution layer. Y^i_h, Y^i_v, Y^i_f, and Y^i_b represent the features after convolution of 1-D transposition convolutions from four directions, which are calculated as follows:

$$\begin{aligned} Y_D[i, j] &= (X * \omega)_D \\ &= \sum_{t=-k}^{k} x[i + D_h t, j + D_w t] \cdot \omega[k - t] \end{aligned} \tag{17}$$

where Y_D is a general term for features Y^i_h, Y^i_v, Y^i_f, and Y^i_b. $[i, j]$ means the value of the pixel at the (i, j) location of the feature. X denotes the input feature. $\omega \in \mathbb{R}^{2k+1}$ denotes a 1-D convolutional filter with a length $2k+1$. $D = (D_h, D_w)$ stands for the direction of the ω filter, where the value of the direction D equals (0,1), (1,0), (1,1), $(-1,1)$ for horizontal, vertical, front diagonal, and back diagonal directions, respectively. $X * \omega$ denotes the convolution calculation. From the parameter analysis of MDSC in the experiments in Section IV, we know that the values of k for the MDSC module in different decoding layers have different influences on the road extraction results. The MDSC module achieves the best experimental results when $[k_1, k_2, k_3, k_4] = [4, 3, 2, 1]$, where k_i indicates the value of the ith decoding layer.

III. Dataset Description

A. Erie Dataset

The study area for Erie Country Data was a region of part of Erie County, PA, USA, including the cities of Buffalo, West Seneca, Cheektowaga, and Lackawanna. The experimental area was approximately 241.55 km^2 and included typical features such as buildings, vehicles, parking lots, forests, rivers, and roads, with urban roads being our objects of study. The Erie data contained both optical imageries and LiDAR data, where the optical imageries were provided by the New York State Digital Ortho imagery Program (which could be downloaded for use from the NYSGIS Clearinghouse website). The optical imageries contained four bands: red band, blue band, green band, and near-infrared band. The downloaded optical imageries had a raw resolution of 0.3048 m (1 foot), for which we used ArcGIS 10.1[1] (Esri, Redlands, CA, USA) for mosaic and relative radiometric correction. The LiDAR data was acquired in the same experimental area and the same shooting year (2008), where the average point spacing was 3.135 feet. LiDAR data was stored in LAS1.0 format, and we used ArcGIS 10.1 software to process it to obtain digital surface model (DSM) data, digital elevation model (DEM) data, and intensity data with a spatial resolution of 1 m. We made the difference between DSM data and DEM data to obtain nDSM data. The nDSM data reflected the height of the ground object relative to the ground, and the

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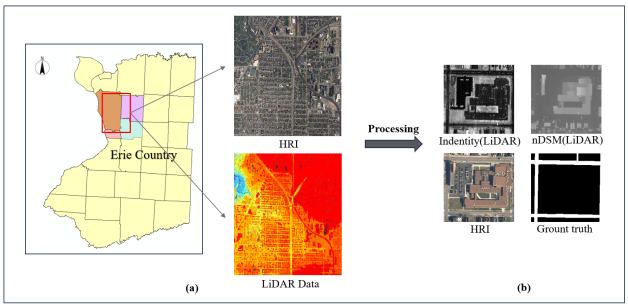

Fig. 4. Overview of the study areas in Erie Road Dataset. (a) Original HRI and LiDAR data of the Erie Country Data. (b) Visual display of images after processing (downsampling and cropping).

intensity data reflected the different material properties of the ground object. The nDSM data and the intensity data together constituted the two characteristic bands of the LiDAR data. In order to fuse the optical image and LiDAR data, we downsampled the optical image to an image with a spatial resolution of 1 m and performed precise relative geometric correction on the high-resolution image and LiDAR data by manually selecting ground control points. The optical images, LiDAR data, and ground truth involved in this study were shown in Fig. 4.

For network training, we made the Erie road dataset by cropping the Erie data without overlapping and randomly dividing the subimages. The Erie road dataset contained training, validation, and test sets, in which the training set could be divided into optical images, LiDAR data, and labels, and the validation set and the test set had similar data divisions, with image sizes of 512×512 and spatial resolutions of 1 m. In order to ensure as much as possible that the road labeling contributions of the training, validation, and test sets were similar, the road dataset was fine-tuned after random splitting. Finally, the training set, validation set, and test set contain 544, 176, and 174 subimages, respectively, with a ratio of about 6:2:2.

B. USGS Dataset

The USGS original data were provided by the U.S. National Geological Survey. The study area for the USGS data included urban or suburban areas of nine cities in the U.S., including Aliton, New York, Atlanta, and others. The experimental area contained typical ground objects such as lakes, buildings, forests, vehicles, and roads. USGS data also contained HRIs and LiDAR data. Most of the optical images contained red, blue, and green bands, and a few images contained near-infrared bands, with an optical resolution of 0.15–0.3 m. Since the LiDAR data of some cities could not be downloaded, and considering the unified training of the model, a total of 16 images from six cities in the original data were finally selected. Among them, the three optical images of Atlanta city had a size of 4000×6000, and the optical images of the remaining cities had a size of 5000×5000. The LiDAR data were stored in LAS format. We used ArcGIS 10.1 software to process the LiDAR data to generate DSM data, DEM data, and intensity data. The nDSM data was obtained from the difference between the DSM data and the DEM data, which together with the intensity data constituted the two characteristic bands of LiDAR data. Since the resolutions of the original optical images were not equal, in order to be able to fuse the optical image and LiDAR data, we downsampled the optical images and LiDAR data appropriately, with the final resolution of 0.3–0.5 m.

To facilitate network training, we made the USGS road dataset by cropping the USGS dataset without overlapping and randomly dividing the submaps and finally fine-tuning it. The USGS road dataset included the training set, the validation set, and the test set, where each of the datasets contained optical images, LiDAR data, and labels. The image size was 512×512 with a spatial resolution of 0.3–0.5 m. The training set, the validation set, and the testing sets contained 472, 156, and 157 subimages, respectively, and their ratio was about 6:2:2.

IV. Experimental Results

In this section, we compare the extraction results of the proposed network model using different backbones and perform the ablation study on the proposed module. To demonstrate the superiority of the proposed model, we compare it with other deep learning models. Among them, U-Net [55] is a network model classically used for semantic segmentation. ResNet [56] optimizes network training using residual learning, which is frequently used as the backbone for many semantic segmentation network models with good results. ConvNeXt [57] is a variant of ResNet that has been shown to infer faster and be more accurate than a similar version of the Swin transformer at the same FLOPs. Consequently, we chose Unet, ResNet18, ResNet34, ResNet50, and ConvNeXt-T as the base framework for comparison. Due to the lack of deep convolutional network models for road extraction based on deep feature fusion of HRI and LiDAR data, we selected the multimodal deep feature fusion networks such as HAFNetE [58], CMGFNet [51], CMFNet [49], and PACSCNet [62] that had been used in building extraction or remote sensing semantic segmentation in recent years as well as ACNet [59], CANet [60], SFAF-MA [61] which had similar network structures for comparison to reflect the superiority of the proposed DECCFNet.

A. Implementation Details

The environment and framework used for the experiments are Python V3.8 and PyTorch 1.10.0, respectively. The CPU is an Intel[2] Core[1] i7-10700F and the GPU is a 24 GB RTX3090. The size of the image fed into the network is 512×512. To improve the generalization of the network, the input data was randomly flipped horizontally and vertically and randomly rotated between $-15°$ and $15°$ for data augmentation. We chose to train the network with an Adam optimizer with an initial learning rate of 0.001 and 150 training epochs and verified by observing that the IOU raised slowly while dynamically updating the learning rate. For the problem of extreme imbalance of the dataset categories, we adopt the

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binary cross entropy and dice coefficient loss as the loss function of the network, and the loss function L can be defined as follows:

$$L_{\text{bce}} = -\frac{1}{N}\sum_{i=1}^{N} y_i \log p_i + (1-y_i)\log(1-p_i) \tag{18}$$

$$L_{\text{dice}} = 1 - \frac{2\times|T\cap P|}{|T|+|P|} \tag{19}$$

$$L = L_{\text{bce}} + \lambda L_{\text{dice}} \tag{20}$$

where y_i denotes the ground truth value of the ith pixel, $y_i = 1$ indicates that the pixel belongs to the road, $y_i = 0$ indicates that the pixel belongs to the background, p_i represents the predicted value of the ith pixel, and N is the total number of pixels. T and P are the ground truth and the predicted value of the road, respectively. According to experience, the value of λ is 1.

B. Evaluation Metric

Consistent with other deep network models for road extraction, we chose intersection over union (IoU), Precision (P), Recall (R), and $F1$-score as evaluation metrics. They are defined as

$$\text{IOU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{21}$$

$$\text{P} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{22}$$

$$\text{R} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{23}$$

$$F1 = 2\times\frac{\text{P}\times\text{R}}{\text{P}+\text{R}} = \frac{2\text{TP}}{2\text{TP}+\text{FP}+\text{FN}} \tag{24}$$

where true positive (TP) is the number of road pixels that were correctly extracted, namely, positive detection, false positive (FP) is the number of pixels that were incorrectly extracted from the background as roads, namely, false detection, true negative (TN) is the number of pixels that were correctly extracted from the background, and false negative (FN) is the number of pixels incorrectly extracted as background missed detection.

C. Hyperparameter Analysis for MDSC Module

The proposed MDSC module was inserted in each decoding layer of the decoder. Each decoding layer $D^i \in \mathbb{R}^{C_i\times H_i\times W_i}$ feature contained different semantic information and sizes, where C_i, H_i, and W_i denoted the number of channel, height, and width for the ith ($i = 1, 2, 3, 4$) decoding layer. We believed that selecting the right size of 1-D convolution for the MDSC module was significant for the network to focus on the road features. To explore the optimal performance of the MDSC module, we set up experiments on the value of k for the 1-D convolutional filters of the MDSC module at different decoding layers. Table I showed the experimental results on the Erie test set. When $k_i = 4$, the filter length of 1-D convolution was 9, which was consistent with the number of parameters of the traditional 3×3 square convolution. The minimum value of k_i was 1. Since in the decoder, the abstraction degree of different decoding layers decreased sequentially, while the high-level semantic features were gradually reduced to the low-level features. Therefore, we tried to decrease k from 4 to 1 to match the degree of semantic information contained in the decoding layer features. The best performance in almost all evaluation metrics was reached when $[k_1, k_2, k_3, k_4] = [4, 3, 2, 1]$, where k_i denoted the value of the ith decoding layer. Therefore, in the next experiments, the MDSC was set to the above value.

TABLE I
HYPERPARAMETRIC ANALYSIS OF THE MDSC MODULE ABOUT THE VALUE OF k_i ON THE ERIE TEST SET

k_i	P↑	R↑	F1↑	IOU↑
[1,1,1,1]	92.37	90.74	91.55	84.41
[2,2,2,2]	**93.05**	90.68	91.85	84.93
[3,3,3,3]	92.63	90.79	91.70	84.67
[4,4,4,4]	92.80	90.70	91.73	84.73
[4,3,2,1]	92.76	**91.19**	**91.96**	**85.12**

TABLE II
QUANTITATIVE COMPARISON OF DIFFERENT BACKBONES OF DECCFNET ON THE ERIE TEST SET

Backbones	P↑	R↑	F1↑	IOU↑
U-Net	92.64	89.87	91.23	83.88
ResNet-18	92.36	90.67	91.50	84.34
ResNet-50	92.01	91.41	91.71	84.68
ConvNeXt-T	**93.57**	89.20	91.33	84.05
ResNet-34	92.76	**91.19**	**91.96**	**85.12**

D. Comparison of Different Backbones in DECCFNet

This section investigated the performance of the proposed DECCFNet based on different backbones. Since the proposed DECCFNet contained two new modules, CMFF and MDSC, which were plug-and-play, the selection of appropriate backbones was also beneficial to improve road extraction accuracy. Table II demonstrated the evaluation metric values of the prediction results of the proposed network when different backbones were employed on the test set of the Erie roads dataset. It should be noted that the backbones of the ResNet series and ConvNeXt-T were loaded with the pretraining weights on the ImageNet dataset in order to accelerate the training of the network, and we replaced the first 7×7 convolutional layer in the initial stage with the Stem block of three 3×3 convolutional layers in order to retain more details and keep the receptive field. It could be seen that ResNet-34 achieved relatively the best road extraction as the backbone of DECCFNet, so we adopt the pretrained ResNet-34 with Stem blocks as the backbone of DECCFNet.

E. Ablation Study

In this section, the ablation study in the proposed DECCFNet demonstrated the effectiveness of the proposed two modules CMFF and MDSC as well as the Stem block replaced in the encoder part for retaining more information. For fair

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TABLE III
ABLATION STUDY FOR DIFFERENT COMPONENTS COMBINATIONS ON THE ERIE TEST SET

Baseline	Stem	CMFF	MDSC	P↑	R↑	F1↑	IOU↑
✓				92.02	88.96	90.47	82.59
✓	✓			91.90	89.88	90.88	83.28
✓	✓	✓		92.74	90.46	91.58	84.47
✓	✓		✓	92.41	90.27	91.33	84.04
✓		✓	✓	92.71	90.77	91.73	84.73
✓	✓	✓	✓	**92.76**	**91.19**	**91.96**	**85.12**

TABLE IV
ABLATION STUDY FOR DIFFERENT COMPONENTS COMBINATIONS ON THE USGS TEST SET

Baseline	Stem	CMFF	MDSC	P↑	R↑	F1↑	IOU↑
✓				93.89	90.40	92.11	85.38
✓	✓			93.46	91.26	92.34	85.78
✓	✓	✓		93.39	92.13	92.75	86.48
✓	✓		✓	93.34	92.29	92.81	86.59
✓		✓	✓	93.67	92.59	93.13	87.14
✓	✓	✓	✓	**94.09**	**92.64**	**93.36**	**87.55**

comparisons, we summed the simple elements of the two modal features as the baseline, then added each module gradually. The performance of each module on the Erie test set was shown in Table III. After replacing the initial 7×7 encoding layer of the baseline with the Stem block of three 3×3 convolutional layers, the IOU result is improved by 0.69%. By adding a Stem block to the CMFF and MDSC modules, the road IOU was improved from 84.73% to 85.12%. The above showed that the Stem block could effectively maintain the respective field size and enhance the nonlinear representation of the network. At the same time, more detailed information was retained, which was beneficial to the subsequent feature extraction. The addition of the CMFF module improved the road IOU from 83.28% to 84.47%, proving that the CMFF could effectively rectify and integrate multimodal features. After the addition of the MDSC module to the decoder, the road IOU improved from 83.28% to 84.04%, which showed that MDSC was indeed able to make the network more focused on road features. Combining the Stem block, CMFF, and MDSC together achieved the best results with an improvement of 2.53% compared to the baseline. Similarly, we did ablation experiments for DECCFNet on the USGS test set. The results of the ablation experiment are shown in Table IV. The addition of Stem blocks, CMFF, and MDSC modules improved the road IOU scores by 0.4%, 0.7%, and 0.81%, respectively. We also compared the results of roads with the addition of a combination of CMFF and MDSC modules on the basis of the Stem block, where the road IOU improved from 85.78% to 87.55%. Finally, the combination of Stem blocks, CMFF, and MDSC modules together also showed the best results on the USGS test set likewise performed the best results. The above emphasized the crucial role played by the diverse components of DECCFNet in obtaining the best results for road extraction.

F. Model Comparison on the Erie Dataset and USGS Dataset

Due to the lack of road extraction networks for deep feature fusion of HRI and LiDAR data modalities, we compared the proposed DECCFNet with recent models that had similar remote sensing ground object extraction methods or similar network structures. HAFNetE [58], CMGFNet [51], CMFNet [49], and PACSCNet [62] were models that were used for multimodal data building extraction or remote sensing semantic segmentation. They did not make specific requirements on the network input data, so we believed that these models could be used to extract roads as well. ACNet [59], CANet [60], and SFAF-MA [61] which had similar network structures to DECCFNet, were multimodal data fusion networks in the field of RGB-D or RGB-T semantic segmentation. Although these semantic models were trained on different data than ours, they also extracted objects from complementary multimodal data, thus we can use the above models for road extraction. Meanwhile, to prove that the proposed model could effectively learn the advantageous complementary information of each modality, we also compared it with the single-branch DECCFNet model trained only with HRI or LiDAR data. To ensure the fairness of the comparison, all models were trained ten times in the same environment to take the average value.

We uniformly evaluated each research network on an RTX3090 graphics card with 24 G memory. We used an image size of 512×512, and the evaluation metrics were FLOPs (G) which represented floating-point operations and Parameters (M). Respectively, the corresponding IoU scores of the models on the Erie dataset and the USGS dataset were given. We also counted the training time of different networks per epoch. The evaluation results were shown in Table V, which demonstrated that our proposed DECCFNet achieved a better balance between efficiency and accuracy than other methods.

Table VI summarized the values of the evaluation metrics for the road prediction results of all models on the Erie test set. It could be seen that the IOUs of the roads extracted by DECCFNet trained with the above two modalities were improved by 2.94% and 3.55%, respectively. Compared with the training with only HIS data or LiDAR data, which proved that DECCFNet could be able to fuse the deep features of each modality effectively. Compared with other models, the P, R, $F1$, and IOU of the proposed network's road evaluation metrics were 92.76%, 91.19%, 91.96%, and 85.12%, respectively, which were the first performance, in which the IOU of the road was improved by at least 0.98%, and the performance indicated that the proposed DECCFNet maintained the superiority and could extract the road features effectively.

We selected the road extraction result maps of the above models on the Erie test set in some complex scene images, as shown in Figs. 5 and 6. Roads obscured by shadows were present in both A and B images in Fig. 5, and it could be seen from the extraction results of Ours (HRI) and Ours (LiDAR) that it was difficult to extract roads obscured by shadows, such as the missed area in the blue part, by relying on HRI data only, while LiDAR data was not affected by shadows. Data fusion after joining LiDAR data in DECCFNet, missed detections of roads became less. Compared with other models,

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TABLE V
PARAMETER ANALYSIS AND TRAINTIME IN PER EPOCH COMPARISON. IMAGE SIZE IS 512 × 512

Method	FLOPs(G)	Parameters(M)	IOU(Erie)	IOU(USGS)	TrainTime(Erie)	TrainTime(USGS)
HAFNetE	**30.09**	**6.98**	83.80	86.28	168s	128s
CMGFNet	152.88	85.22	82.87	85.80	134s	115s
CMFNet	312.09	98.42	83.29	84.24	206s	181s
ACNet	105.47	116.60	84.14	86.51	150s	139s
CANet	110.63	127.28	83.91	86.79	154s	140s
SFAF-MA	111.94	124.88	83.73	86.00	166s	143s
PACSCNet	151.56	89.95	82.62	86.37	164s	142s
Ours	65.47	54.51	**85.12**	**87.55**	**132s**	**113s**

TABLE VI
QUANTITATIVE COMPARISON OF DIFFERENT MODELS ON THE ERIE TEST SET

Method	P↑	R↑	F1↑	IOU↑	Remark
HAFNetE	**93.35**	89.12	91.18	83.80	Multi-modal
CMGFNet	92.76	88.60	90.63	82.87	Multi-modal
CMFNet	91.58	90.19	90.88	83.29	Multi-modal
ACNet	92.77	90.06	91.39	84.14	Multi-modal
CANet	92.57	89.96	91.25	83.91	Multi-modal
SFAF-MA	92.31	90.01	91.14	83.73	Multi-modal
PACSCNet	92.31	88.74	90.49	82.62	Multi-modal
Ours(HRI)	92.55	88.00	90.22	82.18	Single-modal
Ours(LiDAR)	91.20	88.54	89.85	81.57	Single-modal
Ours	92.76	**91.19**	**91.96**	**85.12**	Mulit-modal

our proposed model could effectively fuse HRI data and LiDAR data. Images C, D, and E in Fig. 6 contained complex scenes of railroads, parking lots, building shading, dense roads, and viaducts. As shown in image C, the road near the parking lot, our proposed network could effectively focus on the road features compared to other methods, and the missed part of the road was very little, which proved the effectiveness of the MDSC module. As shown in images D and E, the proposed network was able to distinguish railroads similar to roads and extract complex and dense viaduct roads well compared to other methods, and the red miss-detection part was less. The effectiveness of the CMFF module was also proved.

Similarly, we experimented with road extraction on the USGS dataset. Table VII is a quantitative comparison of the road prediction results for all methods on the USGS test set. From Table VII, we found that DECCFNet also performed superiorly, improving the road IOU scores of the multimodal fusion network DECCFNet by 2.8% and 5.0% compared to the HRI and LiDAR data only. Compared with other models, DECCFNet performed the best in most of the evaluation metrics of road extraction results, with P, R, $F1$, and IOU of 94.09%, 92.64%, 93.36%, and 87.55%, respectively. This indicated the good superiority of the proposed model. Moreover, we selected some road result maps of each model in complex scenarios on the USGS test set, as shown in Fig. 7. Image A in Fig. 7 contained dense roads. B and C images in Fig. 7 presented many roads obscured by trees or buildings, and some railroad interference existed in image D. In the above complex scenarios, the roads obscured by shadows could not be effectively extracted with HRI data only, as shown in the blue missed detections in images B and C. After adding LiDAR data, DECCFNet could effectively extract the obscured roads. Meanwhile, in images A and D, there were more false detections using only LiDAR data, but DECCFNet greatly reduced the false detections of roads after fusing HRI data and LiDAR data. In the above complex sceneries, our proposed network had fewer false and missed road detections compared to other models, which not only proved the effectiveness of the proposed CMFF and MDSC modules but also verified the better generalization of DECCFNet from the visual perspective.

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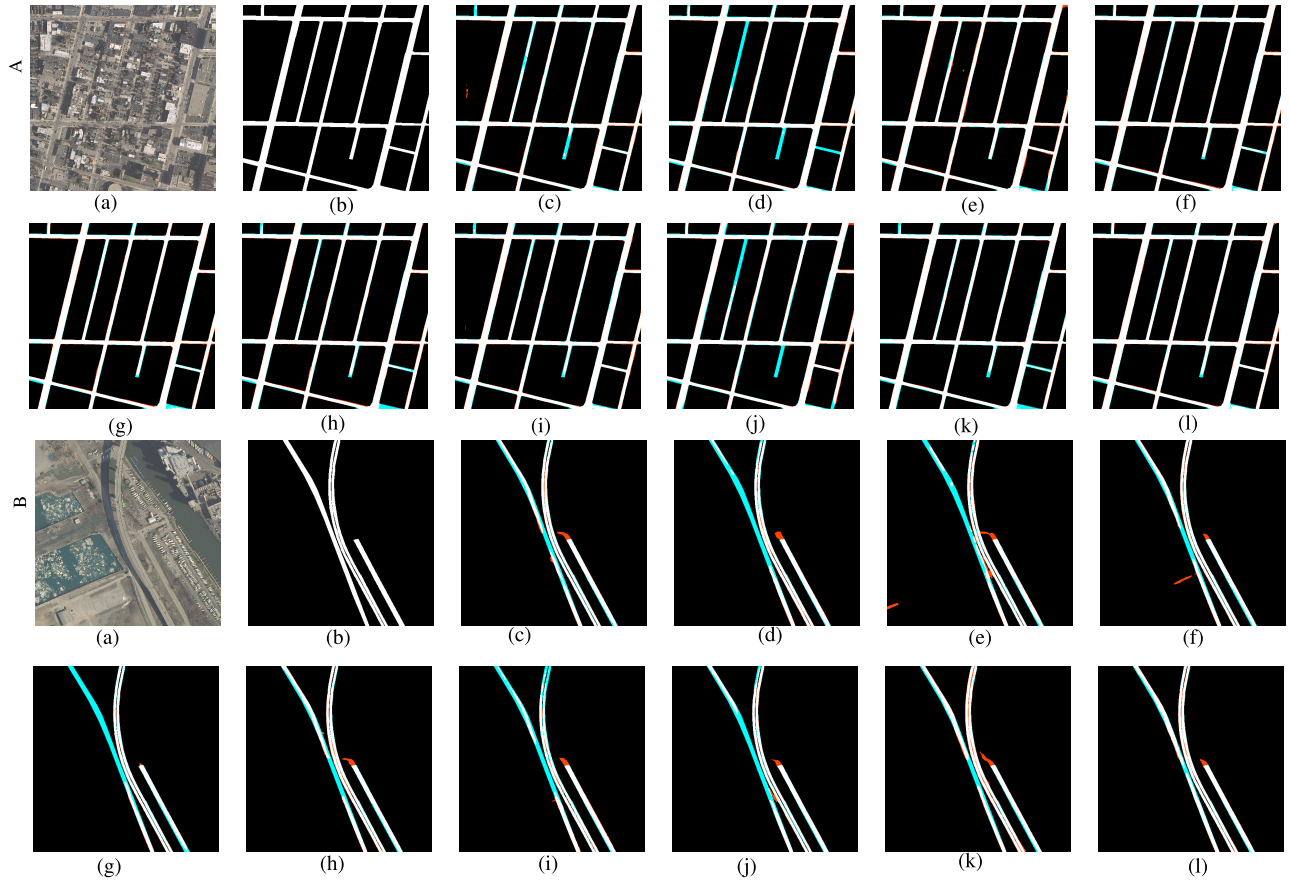

Fig. 5. Visual comparison of the road extraction results of different methods for images A and B on the Erie test set. Black indicates the background, and white, red, and blue indicate positive, false, and missed detection of roads, respectively. Image A: (a) HRI, (b) GT, (c) HAFNetE, (d) CMGFNet, (e) CMFNet, (f) ACNet, (g) CANet, (h) SFAF-MA, (i) PACSCNet, (j) Ours (HRI), (k) Ours (LiDAR), and (l) Ours. Image B: (a) HRI, (b) GT, (c) HAFNetE, (d) CMGFNet, (e) CMFNet, (f) ACNet, (g) CANet, (h) SFAF-MA, (i) PACSCNet, (j) Ours (HRI), (k) Ours (LiDAR), and (l) Ours.

TABLE VII
QUANTITATIVE COMPARISON OF DIFFERENT MODELS ON THE USGS TEST SET

Method	P↑	R↑	F1↑	IOU↑	Remark
HAFNetE	**94.83**	90.54	92.63	86.28	Multi-modal
CMGFNet	93.70	91.05	92.36	85.80	Multi-modal
CMFNet	92.65	90.27	91.45	84.24	Multi-modal
ACNet	93.72	91.83	92.77	86.51	Multi-modal
CANet	93.88	92.00	92.93	86.79	Multi-modal
SFAF-MA	93.23	91.74	92.47	86.00	Multi-modal
PACSCNet	94.36	91.08	92.69	86.37	Multi-modal
Ours(HRI)	92.52	90.98	91.74	84.75	Single-modal
Ours(LiDAR)	90.98	89.91	90.44	82.55	Single-modal
Ours	94.09	**92.64**	**93.36**	**87.55**	Mulit-modal

V. DISCUSSION

Extracting urban roads in complex urban scenes that relied only on single-modal data was frequently challenging. The reasons mainly included 1) roads in HRI could be obscured by buildings and trees with shadow interference, resulting in discontinuity of the extracted roads; and 2) other features that were different from roads would exhibit similar features, thus interfering with the road extraction results. The DECCFNet proposed in this article fully integrated different modal data and focused on road features through the CMFF module and

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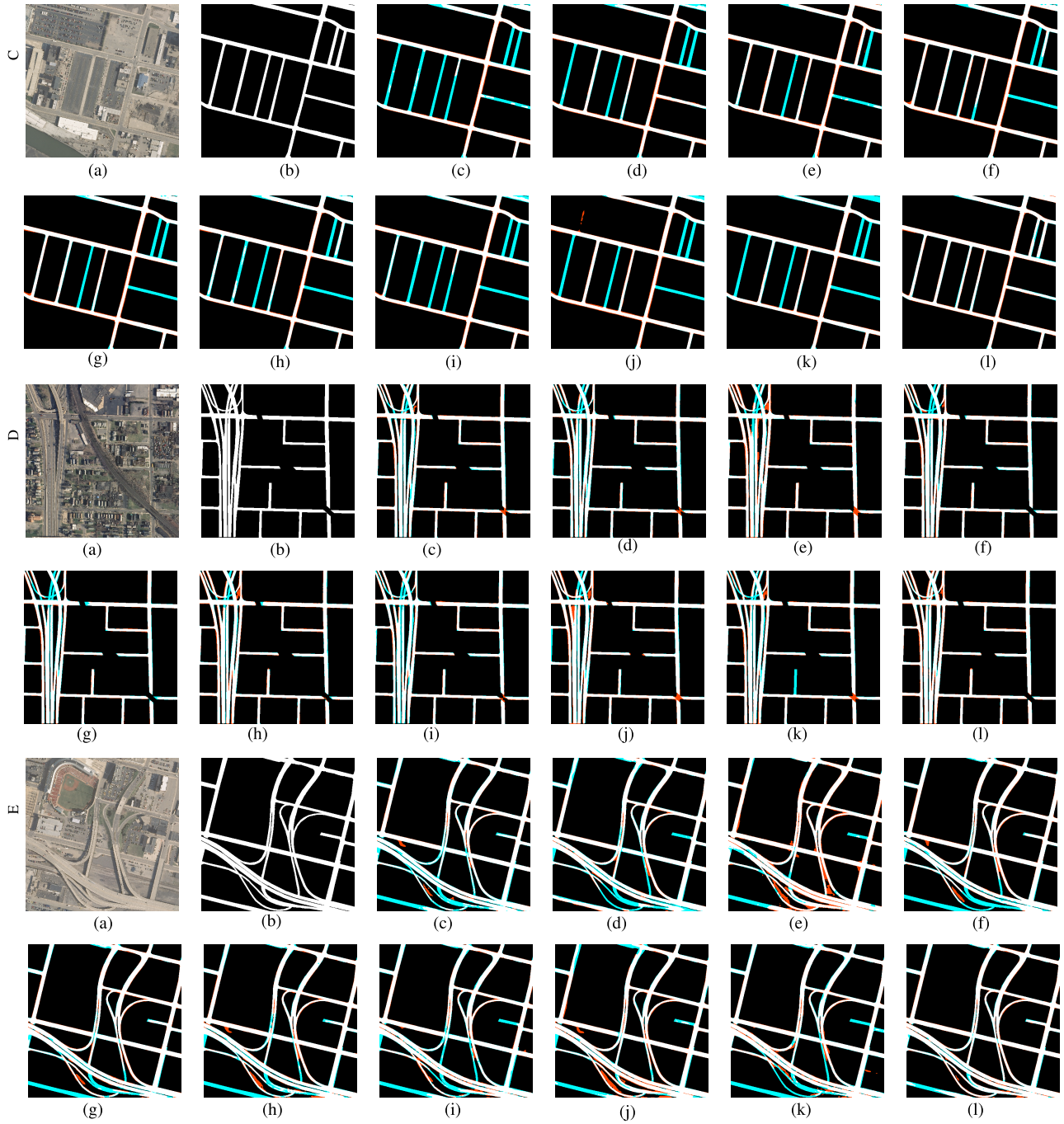

Fig. 6. Visual comparison of the road extraction results of different methods for images C, D, and E on the Erie test set. Black indicates the background, and white, red, and blue indicate positive, false, and missed detection of roads, respectively. Image C: (a) HRI, (b) GT, (c) HAFNetE, (d) CMGFNet, (e) CMFNet, (f) ACNet, (g) CANet, (h) SFAF-MA, (i) PACSCNet, (j) Ours (HRI), (k) Ours (LiDAR), and (l) Ours. Images D: (a) HRI, (b) GT, (c) HAFNetE, (d) CMGFNet, (e) CMFNet, (f) ACNet, (g) CANet, (h) SFAF-MA, (i) PACSCNet, (j) Ours (HRI), (k) Ours (LiDAR), and (l) Ours. Images E: (a) HRI, (b) GT, (c) HAFNetE, (d) CMGFNet, (e) CMFNet, (f) ACNet, (g) CANet, (h) SFAF-MA, (i) PACSCNet, (j) Ours (HRI), (k) Ours (LiDAR), and (l) Ours.

MDSC module, and finally could effectively extract roads and maintain the continuity of roads in complex scenarios, such as those with shadow occlusion, dense roads, and similar road features.

DECCFNet extracted multiscale features for each modality separately through a dual-branch encoder, which contained both shallow features and deep semantic features. We believed that simple summation of different modal features would not only lead to information redundancy but also introduce noise to the superior features of each modal data which caused interference. Therefore, it was necessary to design an effective feature fusion method. The proposed CMFF performed rectified reconstruction and deep fusion of different modal features from the channel and spatial dimension, respectively, using a multiscale feature fusion approach to learn cross-scale semantic information. From Tables III and IV in Section IV-E, we could see the effectiveness of CMFF. On the Erie test set and the USGS test set, the road IOU was improved

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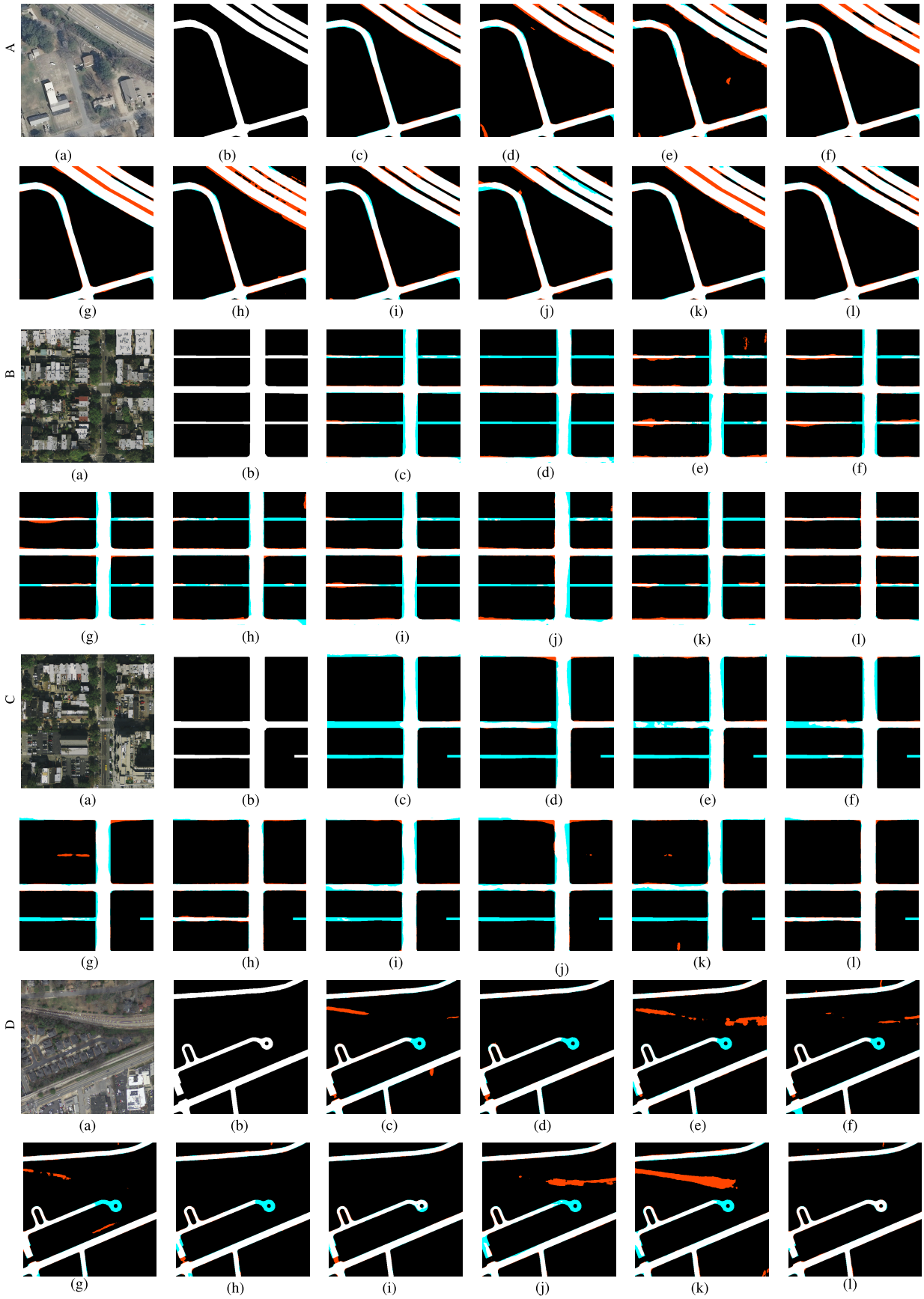

Fig. 7. Visual comparison of the road extraction results of different methods for images A–D on the USGS test set. Black indicates the background, and white, red, and blue indicate positive, false, and missed detection of roads, respectively. Image A: (a) HRI, (b) GT, (c) HAFNetE, (d) CMGFNet, (e) CMFNet, (f) ACNet, (g) CANet, (h) SFAF-MA, (i) PACSCNet, (j) Ours (HRI), (k) Ours (LiDAR), and (l) Ours. Image B: (a) HRI, (b) GT, (c) HAFNetE, (d) CMGFNet, (e) CMFNet, (f) ACNet, (g) CANet, (h) SFAF-MA, (i) PACSCNet, (j) Ours (HRI), (k) Ours (LiDAR), and (l) Ours. Image C: (a) HRI, (b) GT, (c) HAFNetE, (d) CMGFNet, (e) CMFNet, (f) ACNet, (g) CANet, (h) SFAF-MA, (i) PACSCNet, (j) Ours (HRI), (k) Ours (LiDAR), and (l) Ours. Image D: (a) HRI, (b) GT, (c) HAFNetE, (d) CMGFNet, (e) CMFNet, (f) ACNet, (g) CANet, (h) SFAF-MA, (i) PACSCNet, (j) Ours (HRI), (k) Ours (LiDAR), and (l) Ours.

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by 1.19% and 0.7%, respectively. The current convolutional kernel of deep convolutional networks was usually square. Comparatively, roads were narrow and difficult to interrupt compared to other ground objects and required long-distance dependencies and context relationships, which was crucial for road extraction considering these issues. Similarly, the addition of MDSC made the road IOU improve by 0.76% and 0.81% on both datasets, respectively. The results of the ablation study showed that the combination of Stem, CMFF, and MDSC was very powerful, in which the road IOU increased from 82.59% to 85.12% on the Erie test set and from 85.38% to 87.55% on the USGS test set. It is worth noting that the selection of the network backbone had an influential impact on the feature extraction. As we can see from Table II in Section IV-D, the best road extraction results were obtained with ResNet-34 as the backbone, while ResNet-50 and ConvNeXt had higher network complexity compared to ResNet-34, which was less suitable for the Erie road dataset with relatively limited data. Meanwhile, U-Net and ResNet18 had fairly shallow network layers resulting in a weaker representation of the network model that failed to adequately capture complex patterns and relationships. In addition, we experimentally compared the proposed network with networks such as HAFNetE, CMGFNet, and CMFNet on the two test sets, as shown in Tables VI and VII and Figs. 5–7 in Section IV-F. At the same time, we analyzed the corresponding model parameters and training time of the proposed network and other comparison methods. As could be seen from Table V in Section IV-F, our model achieved a good balance between efficiency and accuracy. The proposed network exhibited excellence in terms of both quantitative results and visualization results. The proposed network exhibited excellence in terms of both quantitative results and visualization results. Also, to address the extreme imbalance of road data categories, we adopt a hybrid loss function combining binary cross-entropy loss and Dice loss to train the network, and the experiments demonstrated that DECCFNet achieved the best road extraction results when the value of λ in the hybrid loss function is 1.

VI. Conclusion

Urban road extraction is one of the essential tasks in remote sensing interpretation, which has been widely researched and applied to various aspects of urban planning, traffic management, and automatic navigation. HRI contains rich feature detail information but is susceptible to shadows, while LiDAR data is free from shadow interference but has blurred feature boundaries. In order to solve the problem of how to retain the complementary information between different modalities, a DECCFNet is proposed.

First, we design the CMFF module to rectify and integrate the deep features between different modalities, which breaks through the bottleneck of single-modal data. Simultaneously, for the narrow and hardly interruptible characteristics of roads, the features are convolved from multiple directions by an MDSC module, enabling the proposed network to focus on the road features thus improving the road extraction results. Pretrained ResNet-34 is used as the backbone of the proposed network. Via the ablation experiments and comparison with other multimodal data fusion networks, the following conclusions are drawn: 1) the proposed CMFF and MDSC modules help the network to fuse different modal data with regard to road characteristics, and 2) the proposed network fully learns the complementary information of each modal advantage. Compared to using only HRI data and LiDAR data, the road IOU is improved by 2.94% and 3.55% on the Erie dataset and by 2.8% and 5.0% on the USGS dataset. DECCFNet also shows great advantages compared with the other cross-modal data fusion networks.

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